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PAPER OF PT6-ACC-24CAI0104-B2024-SEM2-CLONE EXAM

Q1. Complex Number

Implements Complex numbers using classes in C++.

The program implements class Complex which gives facility to initialize Complex objects using constructors and add and subtract two Complex objects using + and - operators.

Also add one display() function to display the complex number as **12+4i** format, where 12 is real part and 4 is imaginary part of it.

Sample Input:

2

2

1

1

Sample Output:

3+3i

1+1i

ANSWER:

Test case 1 :

Input:
2
2
1
1
Output:
3+3i
1+1i

Test case 2 :

Input:
12
8
6
1
Output:
18+9i
6+7i

Q2. Rectangle

Write a class called **Rectangle** that represents a rectangular two-dimensional region. Your **Rectangle** objects should have the following methods:

```
public Rectangle(int x, int y, int width, int height)
```

Constructs a new rectangle whose top-left corner is specified by the given coordinates and with the given width and height.

`public int getHeight()`

Returns this rectangle's height.

`public int getWidth()`

Returns this rectangle's width.

`public int getX()`

Returns this rectangle's x-coordinate.

`public int getY()`

Returns this rectangle's y-coordinate.

`public String toString()`

Returns a string representation of this rectangle, such as ***"Rectangle[x=1,y=2,width=3,height=4]".***

Sample Input:

1

2

34

45

Sample Output:

1

2

34

45

Rectangle[x=1,y=2,width=34,height=45]

ANSWER:

Test case 1 :

Input:
1
2
3
4
Output:
1
2
3
4
Rectangle[x=1,y=2,width=3,height=4]

Test case 2 :

Input:
10
11
12
13
Output:
10
11
12
13
Rectangle[x=10,y=11,width=12,height=13]

Q3. Vehicle

You are tasked with designing a C++ program involving hybrid and hierarchical inheritance. Create a set of classes to represent different types of vehicles. Your program should include:

(i) A base class Vehicle with attributes make and year and a method displayDetails() that displays the make and year of the vehicle.

(ii) Two derived classes, Car and Bike, both inheriting from the Vehicle class. The Car class should have an additional attribute, fuelType, and the Bike class should have an additional attribute, engineType. Override the displayDetails() method in both derived classes to display their respective attributes as well.

(iii) Another class, Truck, which inherits from Car. This represents hierarchical inheritance. The Truck class should have an additional attribute, cargoCapacity, and override the displayDetails() method to display all attributes, including those inherited from the Car class.

(iv) Finally, create an instance of each of the Car, Bike, and Truck classes, and call their displayDetails() methods to display the details of these vehicles.

Write the C++ code for the above scenario.

Input Format

First line should contain 3 space separated values representing details of Car: make(string), year (int) and fuel type(string).

Second line should contain 3 space separated values representing details of Bike: make(string), year(int), engine type(string).

Third line should contain 4 space separated values representing details of the truck: make(string), year(int), fuel type(string), cargo capacity(string).

Output Format

Details of Car, Bike and Truck on separate lines as shown in sample test case

Sample Test Case:

Input:

Toyota 2022 Petrol

Honda 2021 Electric

Ford 2020 Diesel 5

Output:

Car Details:

Make: Toyota

Year: 2022

Fuel Type: Petrol

Bike Details:

Make: Honda

Year: 2021

Engine Type: Electric

Truck Details:

Make: Ford

Year: 2020

Fuel Type: Diesel

Cargo Capacity: 5 tons

ANSWER:

Test case 1 :

Input:
BMW 2019 Diesel
Enfield 2021 Petrol
Ford 2020 Diesel 5
Output:
Car Details:
Make: BMW
Year: 2019
Fuel Type: Diesel

Bike Details:
Make: Enfield
Year: 2021
Engine Type: Petrol

Truck Details:
Make: Ford
Year: 2020
Fuel Type: Diesel
Cargo Capacity: 5 tons

Test case 2 :

Input:
Nissan 2017 Diesel
Harley-Davidson 2020 Gasoline
Toyota 2015 Petrol 8
Output:
Car Details:
Make: Nissan
Year: 2017
Fuel Type: Diesel

Bike Details:
Make: Harley-Davidson
Year: 2020
Engine Type: Gasoline

Truck Details:
Make: Toyota
Year: 2015
Fuel Type: Petrol
Cargo Capacity: 8 tons

Q4. Date

Write a class named **Date** that remembers information about a month and day. Ignore leap years and don't store the year in your object. You must include the following public methods:

```
Member-name      Description
Date(int m, int d)       constructs a new date representing the given month and day
daysInMonth()    returns the number of days in the month stored by your date object
getDay()         returns the day
getMonth()       returns the month
nextDay()        advances the Date to the next day, wrapping to the next month and/or year if necessary
toString()       returns a string representation e.g. If date is 27 June then return "6/27"
absoluteDay()    return the absolute day of a particular date e.g. January 1 is 1, February 1 is 32, March 1 is 60 and December 31 is 365.
```

You should define the entire class including the class heading, the private fields, and the declarations and definitions of all the public methods and constructor.

Sample Input:

1 \\month

1 \\day

Sample Output:

1 //absolute day

31 //days in month

1 //day

1 //month

2 //next day

1 // month of next day

1/2 // string month/day

ANSWER:

Test case 1 :

Input:
1
1
Output:
1
31
1
1
2
1
1/2

Test case 2 :

Input:
1
4
Output:
4
31
4
1
5
1
1/5

Q5. Bank Account

In a bank, different customers have savings account. Some customers may have taken a loan from the bank. So bank always maintains information about bank depositors and borrowers. Design a Base class Customer (name, phone-number). Derive a class Depositor(accno, balance) from Customer. Again, derive a class Borrower (loan-no, loan-amt) from Depositor. Write necessary member functions to read and display the details of 'n' customers.

Sample

Sample Input:

2 // Number of customers

riya // 1st customer name

98764345 // mobile number

34 // A/C No.

56000 // Balance Amount

32 // Loan number

700000 //Loan Amount

siya // 2nd customer name

6765487878 // customer phone number

45 // A/C No.

3000 // Balance Amount

5 // Loan no

200000 // Loan Amount

Output:

customer details

riya

98764345

34

56000

32

700000

customer details

siya

6765487878

45

3000

5

200000

ANSWER:

Test case 1 :

Input:
1
ram
989876
23
2000
21
400000

Output:
customer details
ram
989876
23
2000
21
400000

Test case 2 :

Input:
2
riya
987
34
56000
32

700000
siya
67654
45
3000
5
200000
Output:
customer details
riya
987
34
56000
32
700000
customer details
siya
67654
45
3000
5
200000

Q6. Polymorphism - Area

Calculate the volume of geometrical shapes using virtual functions for below shapes. Use the following formulas and display the volume upto 3 decimal places.

1. **Volume of sphere= 4/3*3.14*radius*radius*radius**
2. **Volume of cylinder=3.14*radius*radius*height**
3. **Volume of cone=1/3*3.14*radius*radius*height**
4. **Volume of cube= radius*radius*radius**

Choice 1 for sphere, 2 for cylinder, 3 for cone, 4 for cube

Sample input 1

```
1  //choice for sphere
3  //radius of sphere
```

Sample output 1

```
Volume of sphere is 112.757
```

Sample input 2

```
2  //choice for cylinder
3 //radius of cylinder
4 //height of cylinder
```

Sample output 2

```
Volume of cylinder is 113.040
```

Sample input 3

```
5 // wrong choice
```

Sample output 3

```
wrong choice
```

ANSWER:

Test case 1 :

Input:
2
4
5
Output:
Volume of cylinder is 251.200

Test case 2 :

Input:
8
Output:
wrong choice

Q7. Person, Professor and Student

In a university's academic management system, you are tasked with implementing a class hierarchy to represent both Professors and Students, inheriting from the base class Person. The classes should maintain information about age, unique identification numbers (IDs), and other specific attributes.

For this purpose-

Create three classes **Person, Professor** and **Student**.

- The class Person should have data member age.
- The classes Professor and Student should inherit from the class Person.
- The class Professor should have two integer members: publications and cur_id. There will be two member functions: getdata and putdata. The function getdata should get the input from the user: the age and publications of the professor. The function putdata should print the age, publications and the cur_id of the professor.
- The class Student should have two data members: marks, which is an array of size 6 and cur_id. It has two member functions: getdata and putdata. The function getdata should get the input from the user: the age, and the marks of the student in subjects. The function putdata should print the age, sum of the marks and the cur_id of the student.
- For each object being created of the Professor or the Student class, sequential id's should be assigned to them starting from 1.

Your solution should show proper encapsulation, inheritance, and usage of constructors, virtual functions, and static variables.

Sample Input

```
2  //Number of persons
1  // choice 1 for
56 99 //age and publications of professor
2   // choice 2 for student
18 50 48 97 76 34 98   //age and marks of 6 subjects of student
```

Sample Output

```
56 99 1  // age, publications and the cur_id of the professor.
18 403 1  //age, sum of the marks and the cur_id of the student.
```

ANSWER:

Test case 1 :

Input:
2
2
22 10 12 0 18 45 50
1
58 87
Output:
22 135 1
58 87 1

Test case 2 :

Input:
2
2
18 50 48 97 76 34 98

1
78 34
Output:
18 403 1
78 34 1